



STATS/CSE 780
Technical Supplemental

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```
# Load Bank Marketing Archive dataset
```

```
original_data<- read_csv("CC_GENERAL.csv",show_col_types = FALSE)
```

```
# Summary of Data
```

```
dim(original_data)
```

```
## [1] 8950 18
```

```
#summary(original_data)
```

```
cat("Number of missing values is",sum(is.na(original_data)),  
    "\n")
```

```
## Number of missing values is 314
```

```
clean_data <- na.omit(original_data)
```

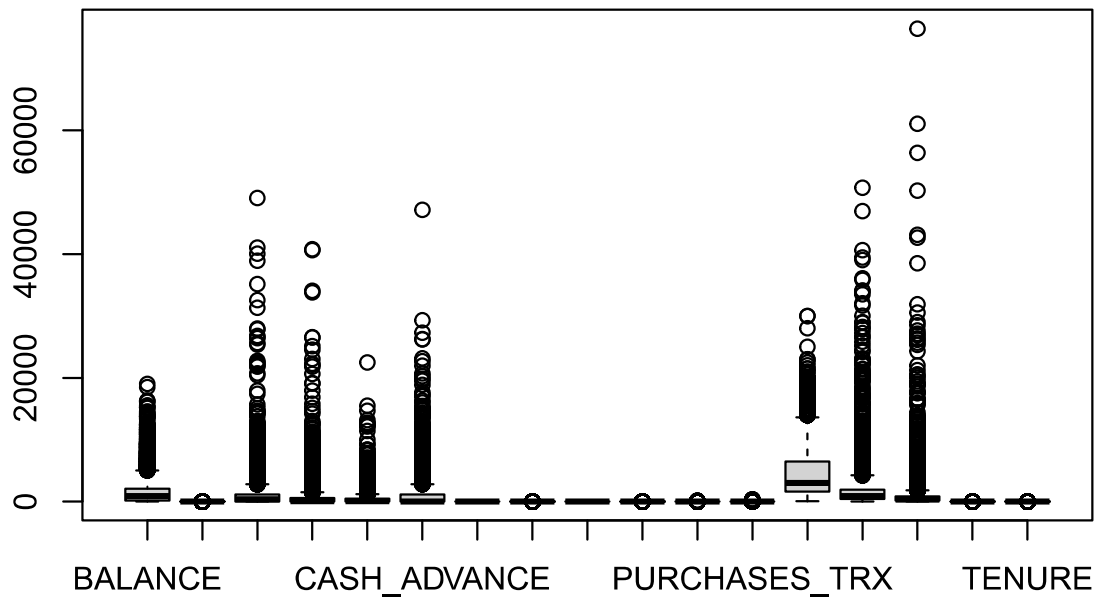
```
# check the missing values
```

```
sum(is.na(clean_data))
```

```
## [1] 0
```

```
clean_data <-clean_data[,-1]
```

```
boxplot(clean_data)
```



```
# Extract column names
c <- colnames(original_data[,-1])

# Define function for calculating IQR and removing outliers
IQRr <- function(x){
  Q1 <- quantile(x, 0.25)
  Q3 <- quantile(x, 0.75)
  IQR <- Q3 - Q1
  lower_bound <- Q1 - 1.5 * IQR
  upper_bound <- Q3 + 1.5 * IQR
  x[x < lower_bound] <- NA
  x[x > upper_bound] <- NA
  return(x)
}
```

```

}

# Apply function to all columns
iqr_list <- lapply(clean_data[, c], IQR)
iqr_df <- as.data.frame(iqr_list)
clean_data <- na.omit(iqr_df)

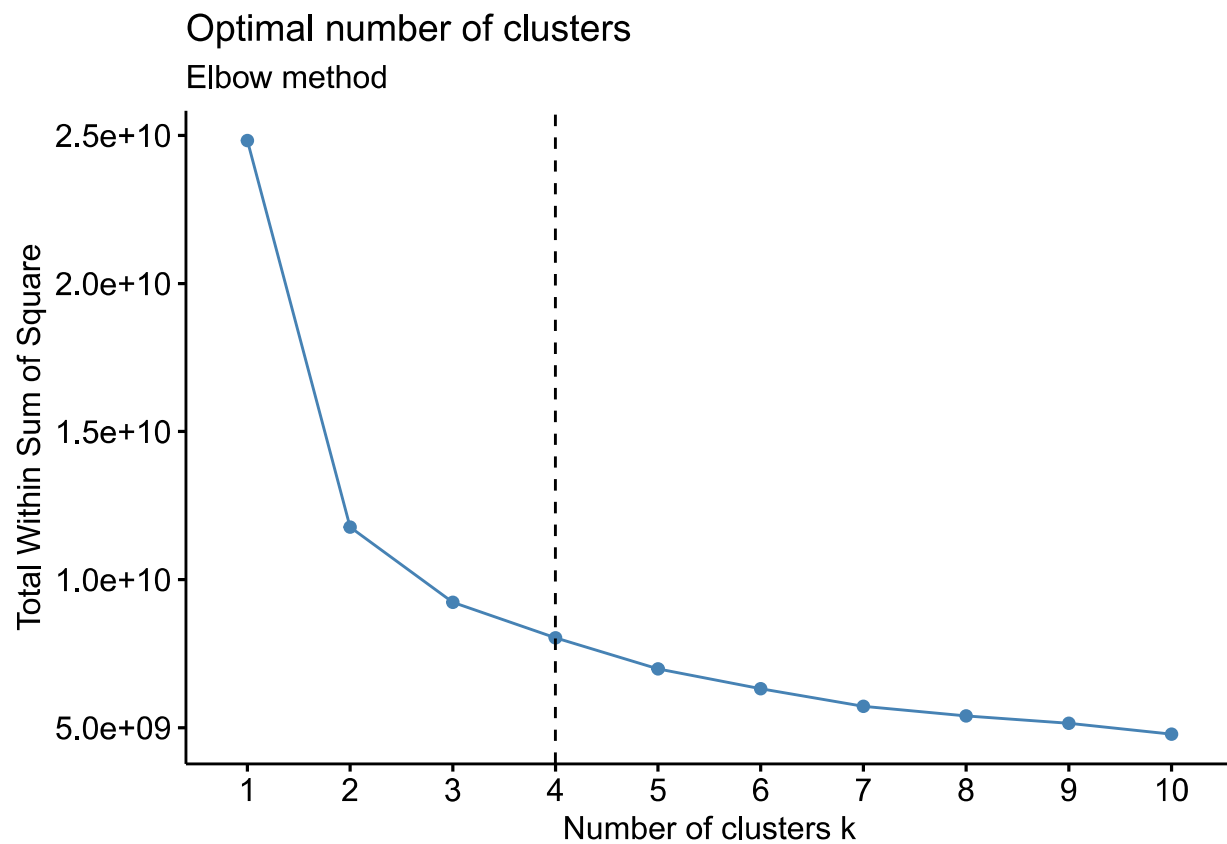
```

Elbow method

```

fviz_nbclust(clean_data, FUNcluster = kmeans, method = "wss") +
  geom_vline(xintercept = 4, linetype = 2) +
  labs(subtitle = "Elbow method")

```

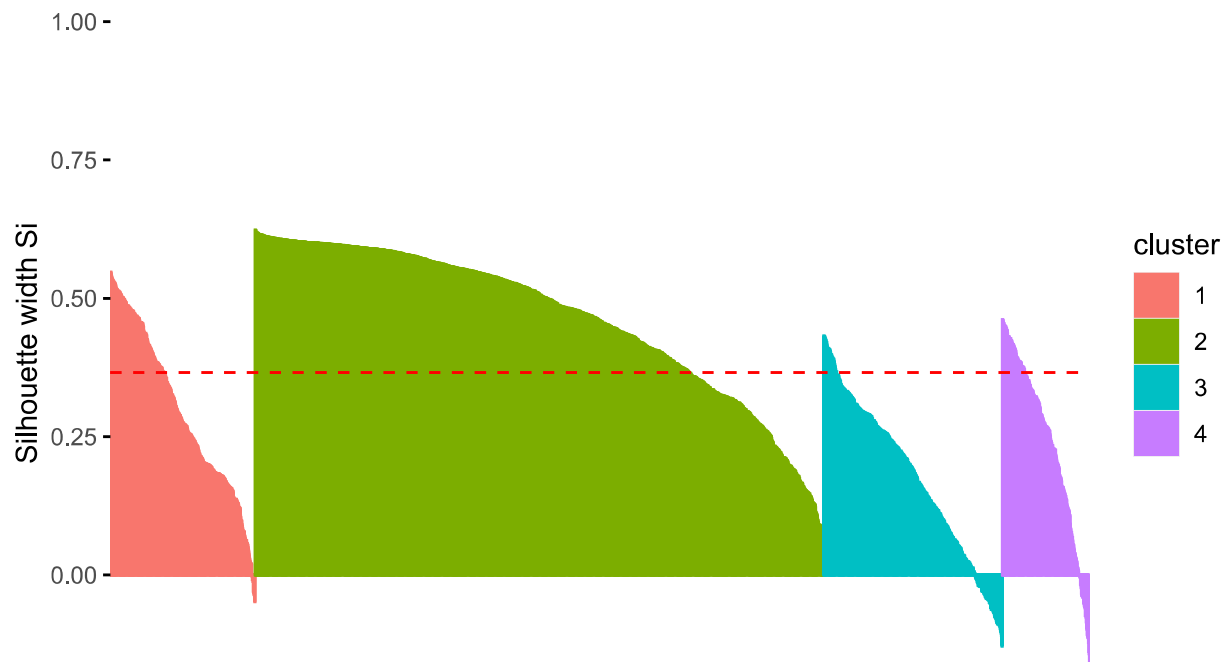


```
set.seed(2)
clustering <- kmeans(clean_data, centers = 4, nstart = 20)
```

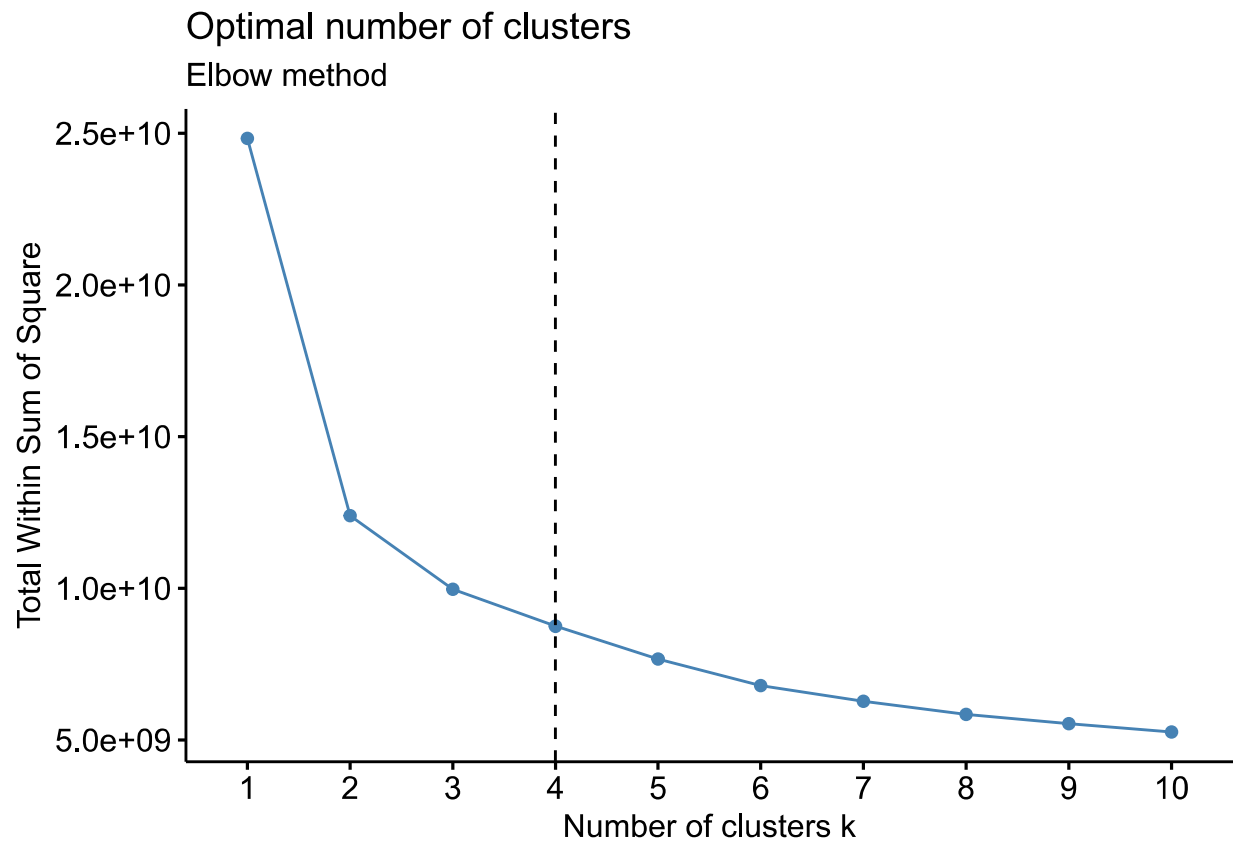
```
sil <- silhouette(clustering$cluster, dist(clean_data))
fviz_silhouette(sil)
```

```
##   cluster size ave.sil.width
## 1         1  436         0.29
## 2         2 1714         0.46
## 3         3  538         0.17
## 4         4  263         0.24
```

Clusters silhouette plot
Average silhouette width: 0.37



```
fviz_nbclust(clean_data, FUNcluster = hcut, method = "wss") +
  geom_vline(xintercept = 4, linetype = 2) +
  labs(subtitle = "Elbow method")
```



the best cutoff in this clustering is 4.

```
# Set cluster number
m <- 4

# Set cluster name
c1 <- "cluster1"
c2 <- "cluster2"
c3 <- "cluster3"
c4 <- "cluster4"

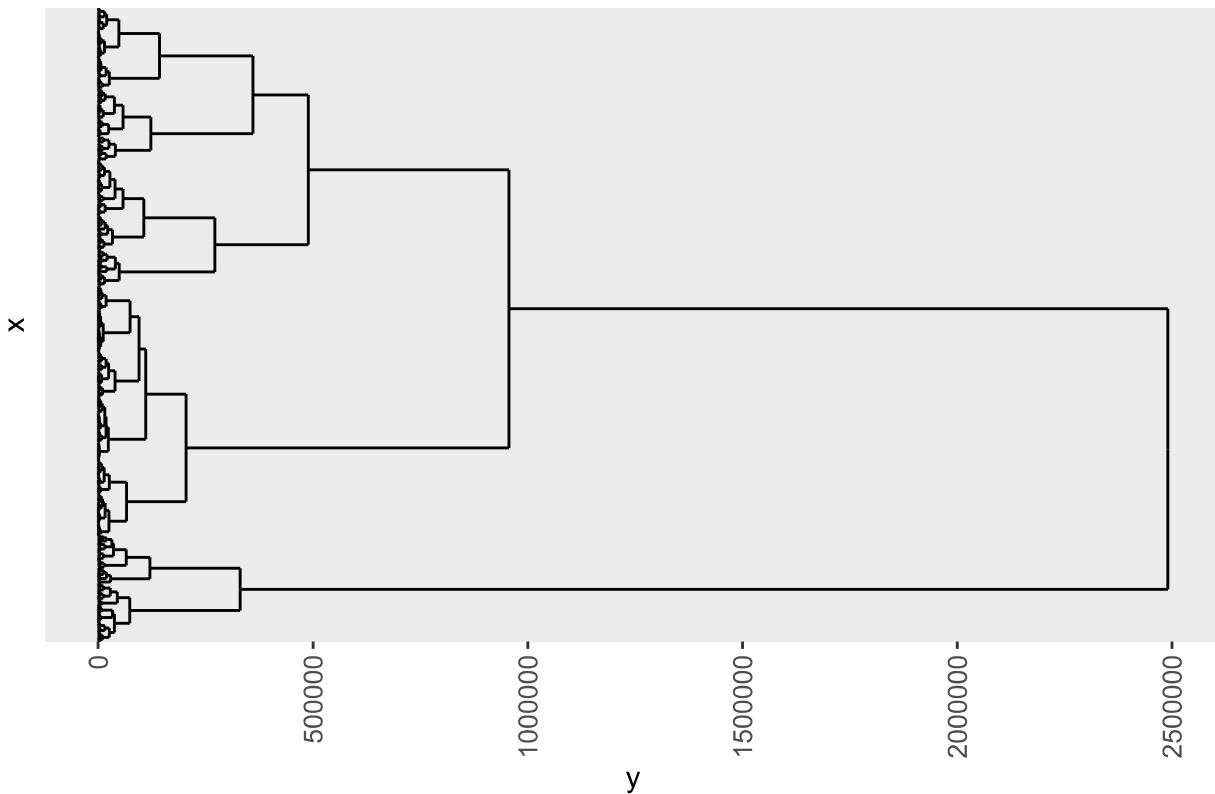
# Perform hierarchical clustering
cluster <- clean_data %>%
  dist(method = "euclidean") %>%
  hclust(method = "ward.D")
```

```
# Plot dendrogram
ggdendrogram(cluster, rotate = TRUE, theme_dendro = FALSE) +
  labs(title = paste0("Hierarchical clustering with ", m, " clusters")) +
  scale_x_continuous(expand = c(0, 0)) +
  theme(panel.grid.minor = element_blank(),
        panel.grid.major = element_blank(),
        axis.text.y = element_blank(),
        axis.ticks.y = element_blank(),
        plot.title = element_text(size = 14, face = "bold"),
        axis.text.x = element_text(angle = 90, hjust = 1, vjust = 0.5, size = 10))
```

Scale for x is already present.

Adding another scale for x, which will replace the existing scale.

Hierarchical clustering with 4 clusters



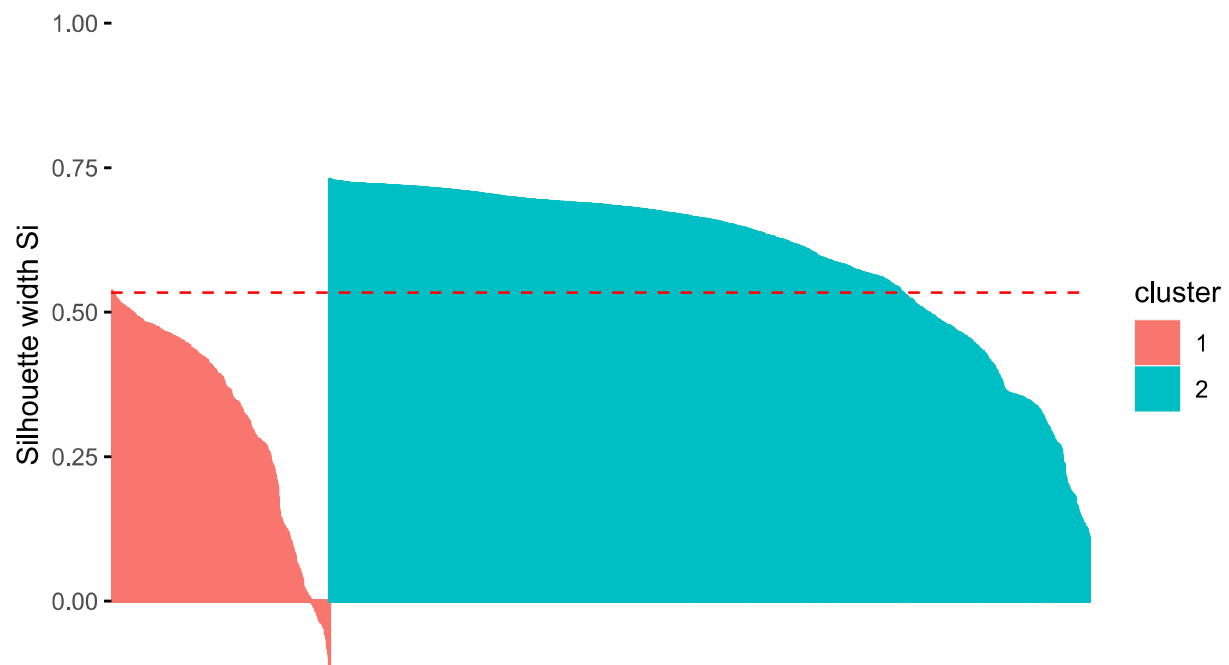
```
pca.out <- prcomp(clean_data, scale. = FALSE)

set.seed(123)

km.out <- kmeans(pca.out$x[,1:2], 2, nstart = 20)
km_pca <- km.out
sil <- silhouette(km_pca$cluster, dist(clean_data))
fviz_silhouette(sil)
```

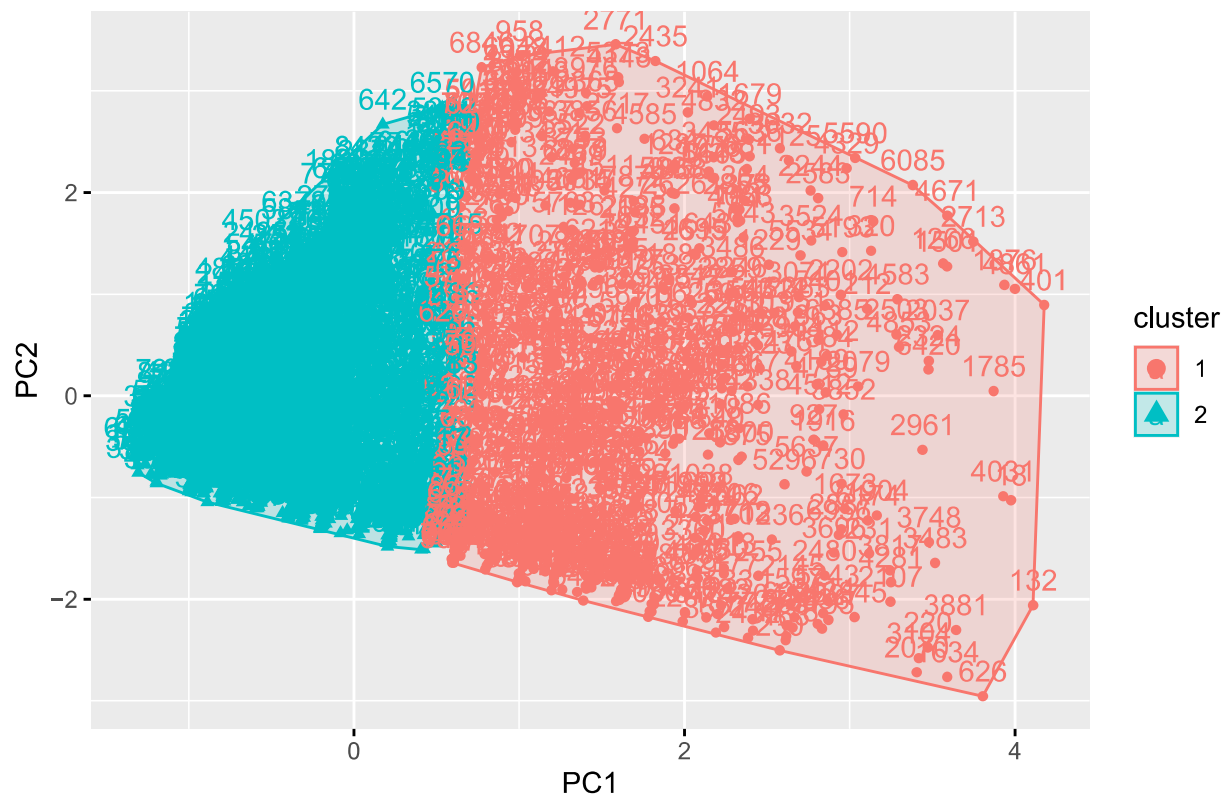
```
##   cluster size ave.sil.width
## 1         1  658          0.32
## 2         2 2293          0.60
```

Clusters silhouette plot
Average silhouette width: 0.53



```
fviz_cluster(km.out, pca.out$x[,1:2])
```

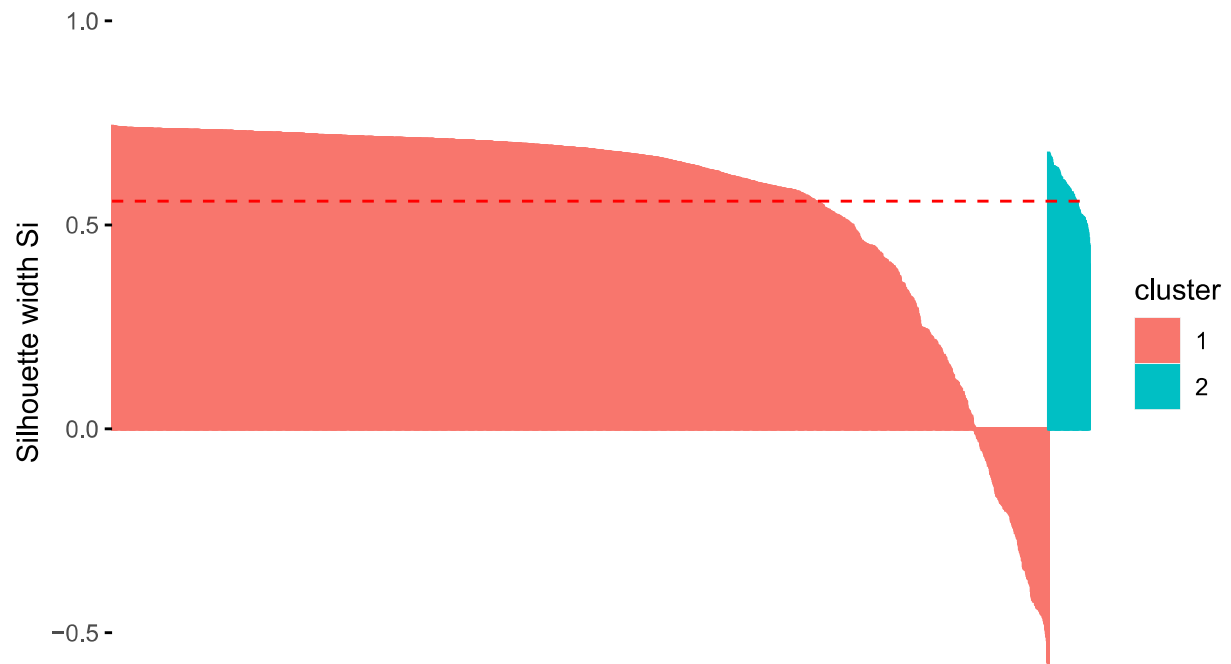

Cluster plot



```
hc.out <- prcomp(clean_data, scale. = FALSE)
hc_pca <- hclust(dist(pca.out$x[,1:2]))
hc_clusters <- cutree(hc_pca, k=2)
sil <- silhouette(hc_clusters, dist(clean_data))
fviz_silhouette(sil)
```

```
##   cluster size ave.sil.width
## 1      1 2824          0.56
## 2      2  127          0.58
```

Clusters silhouette plot
Average silhouette width: 0.56



```
fviz_cluster(list(data = pca.out$x[,1:2], cluster = hc_clusters))
```

